

Veera Manikandan A

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github.com/veeramani27 | linkedin.com/in/veeramani27 | leetcode.com/Veera_Manikandan_A

PORTFOLIO : veeramani27.github.io/portfolio

Education

Sri Venkateswara College of Engineering
B.E. Computer Science and Engineering

2022 – 2026
CGPA: 8.68

Professional Experience

Genworx

Aug 2025 – Feb 2026

AI & Software Development Trainee

- Developed full-stack applications with modular microservices architecture using **FastAPI** and **React** with strict adherence to **SOLID principles** and **JWT/RBAC** security.
- Built AI-driven multi-agent workflows using **LangChain** and **LangGraph**, integrating RAG with vector embeddings and a vector database to automate complex tasks.
- Optimized PostgreSQL performance via execution plan analysis and strategic indexing, reducing average API latency by **35%** in local load tests.
- Enforced code quality with **PyTest unit tests** (85% coverage) and Git-based code reviews, practicing Agile development.

Arka Automaations Limited

June 2024 – July 2024

IoT & Software Development Intern

- Built a real-time IoT monitoring system using ESP32 (C++) and a React dashboard, enabling remote visualization of telemetry data and device control over Internet.
- Designed an event-driven communication pipeline using the **MQTT protocol**, allowing asynchronous data exchange between devices and services with **sub-500ms** latency.
- Engineered a backend service to bridge MQTT event streams with **WebSockets**, streaming real-time telemetry to the dashboard and asynchronously routing user commands to devices via MQTT, with fault-tolerant retry and reconnection mechanisms.

Projects

LexiAI: Multi-Agent Legal Research Engine

Python, LangGraph, FastAPI, ChromaDB, React, Semantic Search, RAG, Tailwind CSS

- Architected a **Multi-Agent RAG System** using **LangGraph**, orchestrating a specialized **Generator Agent** (LLM) and a **Discriminator Agent** to validate responses, significantly reducing hallucination rates and enabling web-based precedent matching for enhanced legal research.
- Implemented **Semantic Search** using vector embeddings (SentenceTransformers) over a corpus of 200+ legal documents, enabling context-aware retrieval beyond keyword matching.
- Designed an asynchronous ingestion pipeline using **asyncio** to parse and embed PDF documents concurrently, optimizing throughput for large datasets.
- Built a RESTful API layer with exponential retry logic handling multiple concurrent web requests.

High-Concurrency Movie Booking System

Python, FastAPI, PostgreSQL, React, SQL Alchemy, Pessimistic Locking, Tailwind CSS

- Solved the **"Double Booking" race condition** by implementing a **Pessimistic Locking strategy** using a dedicated reservation table, ensuring strict **ACID compliance** during concurrent seat requests.
- Designed a normalized database schema (3NF) with strategic indexing on search columns, optimizing query performance for high-frequency seat availability checks.
- Simulated high-load scenarios using Locust, validating the locking mechanism under **500+ concurrent transactions** with zero data inconsistencies.
- Implemented a secure authentication flow with **JWT Access/Refresh tokens**.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Backend Development: FastAPI, Node.js, Express.js, ORMs, REST APIs, JWT Authentication

AI & Data: LangChain, LangGraph, Vector Embeddings, Semantic Search, Chunking

Frontend Development: React (Web & Native), HTML, CSS, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases

Tools & Platforms: Git, GitHub, Docker, Linux, PyTest

Achievements

- Solved **525+ DSA problems** in LeetCode (275+ Medium, 28+ Hard) with a contest rating of 1,848.
- GATE 2026 qualified with a score of 510 and secured AIR 6956 out of 211020 applicants.
- Organized department-level paper presentation and Azure Data Engineering workshop (Interrupt '25).
- Completed Web Development Bootcamp on Udemy, gaining hands-on experience in building full-stack web applications.